

# math-blurbs

## self-adjoint

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This project is hosted on GitHub: <https://github.com/self-adjoint/math-blurbs>

## 0 Introduction

### 0.1 The QED Symbol

**QED** stands for *quod erat demonstrandum* (Latin for “that which was to be demonstrated”), and is used to signify the end of a mathematical proof. Most mathematicians use either a white square  $\square$  or a black square  $\blacksquare$  for this. **Paul J., Sally, Jr.** (1933 – 2013) famously used a sketch of himself in an eyepatch<sup>1</sup> as the QED symbol in his books.

**Theorem 0.1.** For any group of people, there are at least two people with the same number of friends.

**Proof.** Suppose there are  $n$  people in the group. Each person can have any number of friends from 0 to  $n - 1$  (so there are  $n$  possible values). However, there cannot be both a person with 0 friends (friends with no one) and a person with  $n - 1$  friends (friends with everyone). Thus, there must be at least two people with the same number of friends. 😎

The QED symbol I have chosen to use is the **Smiling Face with Sunglasses** emoji<sup>2</sup>. This version has been used by **WhatsApp** since 2017.

**Theorem 0.2** (Cool thing). This is a theorem

**Theorem 0.2**

<sup>1</sup>He was legally blind in his left eye, and nicknamed “The Math Pirate” and “Professor Pirate” by his students.

<sup>2</sup>Not to be confused with an **emoticon**, which refers to things like :) and :( and were mainly used before emojis were introduced into **Unicode 6.0**.

# 1 Limit inferior and limit superior

**Definition 1.1.** Suppose  $(a_n)_{n=1}^{\infty}$  is a sequence in  $\mathbb{R}$ . The **limit inferior** and **limit superior** of  $(a_n)_{n=1}^{\infty}$  respectively are given by:

$$\liminf_{n \rightarrow \infty} a_n = \sup_{n \in \mathbb{N}} \inf_{k \geq n} a_k \qquad \limsup_{n \rightarrow \infty} a_n = \inf_{n \in \mathbb{N}} \sup_{k \geq n} a_k$$

Note that  $\liminf_{n \rightarrow \infty} a_n$  and  $\limsup_{n \rightarrow \infty} a_n$  are allowed to be  $\pm\infty$ .

## 1.1

Suppose  $(a_n)_{n=1}^{\infty}$  is a sequence in  $\mathbb{R}$  and  $x \in \mathbb{R}$ . Prove that:

1.  $x < \liminf_{n \rightarrow \infty} a_n$  if and only if  $x < a_n$  for all but finitely many  $n \in \mathbb{N}$ .
2.  $x < \limsup_{n \rightarrow \infty} a_n$  if and only if  $x < a_n$  for infinitely many  $n \in \mathbb{N}$ .

Do these statements hold if we replace the strict inequalities with non-strict inequalities? Prove or give a counterexample for each one.

## 1.2

Suppose  $(a_n)_{n=1}^{\infty}$  and  $(b_n)_{n=1}^{\infty}$  are sequences in  $\mathbb{R}$ . Prove that:

$$\liminf_{n \rightarrow \infty} a_n + \liminf_{n \rightarrow \infty} b_n \leq \liminf_{n \rightarrow \infty} (a_n + b_n) \leq \liminf_{n \rightarrow \infty} a_n + \limsup_{n \rightarrow \infty} b_n \leq \limsup_{n \rightarrow \infty} (a_n + b_n) \leq \limsup_{n \rightarrow \infty} a_n + \limsup_{n \rightarrow \infty} b_n$$

For each inequality, give an example of sequences  $(a_n)_{n=1}^{\infty}$  and  $(b_n)_{n=1}^{\infty}$  where it is strict. Can you find a single example of  $(a_n)_{n=1}^{\infty}$  and  $(b_n)_{n=1}^{\infty}$  where *all four* inequalities are strict?

**Definition 1.2.** Suppose  $(A_n)_{n=1}^{\infty}$  is a sequence of subsets of  $\mathbb{R}$ . The **limit inferior** and **limit superior** of  $(A_n)_{n=1}^{\infty}$  respectively are given by:

$$\liminf_{n \rightarrow \infty} A_n = \bigcup_{n=1}^{\infty} \bigcap_{k \geq n} A_k \qquad \limsup_{n \rightarrow \infty} A_n = \bigcap_{n=1}^{\infty} \bigcup_{k \geq n} A_k$$

## 1.3

Suppose  $(A_n)_{n=1}^{\infty}$  is a sequence of subsets of  $\mathbb{R}$  and  $x \in \mathbb{R}$ . Prove that:

1.  $x \in \liminf_{n \rightarrow \infty} A_n$  if and only if  $x \in A_n$  for all but finitely many  $n \in \mathbb{N}$ .
2.  $x \in \limsup_{n \rightarrow \infty} A_n$  if and only if  $x \in A_n$  for infinitely many  $n \in \mathbb{N}$ .

## 1.4

State and prove an analog of **Exercise 1.2** for sequences of subsets of  $\mathbb{R}$ .

**Definition 1.3.** A function  $f : \mathbb{R} \rightarrow \mathbb{R}$  is **additive** if  $f(x+y) = f(x) + f(y)$  for all  $x, y \in \mathbb{R}$ .

**Remarks.**

1. The equation  $f(x+y) = f(x) + f(y)$  is known as the *Cauchy functional equation*.
2. It follows by replacing  $x$  with  $x-y$  that  $f(x-y) = f(x) - f(y)$  for all  $x, y \in \mathbb{R}$ . It then follows by substituting  $x=y$  that  $f(0) = 0$ .

**Lemma 1.4.** Suppose  $f : \mathbb{R} \rightarrow \mathbb{R}$  is additive. Then  $f(cx) = cf(x)$  for all  $c \in \mathbb{Q}$  and all  $x \in \mathbb{R}$ .

**Proof.** Suppose  $x \in \mathbb{R}$ . We first show that  $f(nx) = nf(x)$  for all  $n \in \mathbb{Z}$ . Clearly this holds for  $n = 1$  as both sides reduce to  $f(x)$ . Suppose the equation holds for some  $n \in \mathbb{Z}$ . Then we have:

$$\begin{aligned} f((n+1)x) &= f(nx+x) = f(nx) + f(x) = nf(x) + f(x) = (n+1)f(x) \\ f((n-1)x) &= f(nx-x) = f(nx) - f(x) = nf(x) - f(x) = (n-1)f(x) \end{aligned}$$

Thus the equation holds for  $n+1$  and  $n-1$ , and so it holds for all  $n \in \mathbb{Z}$ .

Now suppose  $c \in \mathbb{Q}$ . Then  $c = \frac{p}{q}$  for some  $p, q \in \mathbb{Z}$ ,  $q \neq 0$ . By what we have just shown, we have  $f(px) = pf(x)$  and  $f(qy) = qf(y)$  for all  $x, y \in \mathbb{R}$ . Substituting  $y = \frac{x}{q}$  yields  $f(x) = qf(\frac{x}{q})$ , i.e.  $f(\frac{x}{q}) = \frac{1}{q}f(x)$ , and so  $f(\frac{p}{q}x) = \frac{p}{q}f(x)$ .

Thus  $f(cx) = cf(x)$  for all  $c \in \mathbb{Q}$  and all  $x \in \mathbb{R}$ . 🧐

**Proposition 1.5.** If  $f : \mathbb{R} \rightarrow \mathbb{R}$  is additive and Lebesgue measurable, then  $f$  is continuous at 0.

**Proof.** Suppose  $U \subseteq \mathbb{R}$  is an open set containing 0 and  $V \subseteq U$  is an open set such that  $\{x - y \mid x, y \in V\} \subseteq U$ . Also suppose  $(q_n)_{n=1}^\infty$  is an enumeration of  $\mathbb{Q}$ . Then we have  $\bigcup_{n=1}^\infty (q_n + V) = \mathbb{R}$ , and so  $\bigcup_{n=1}^\infty f^{-1}(q_n + V) = \mathbb{R}$  as well. For each  $n \in \mathbb{N}$ , define  $W_n = f^{-1}(q_n + V)$ . Then we have  $\bigcup_{n=1}^\infty W_n = \mathbb{R}$ , so  $\mu(W_n) > 0$  for at least one  $n \in \mathbb{N}$ . By Sierpinski's theorem on sets of differences, the set  $\{x - y \mid x, y \in W_n\}$  contains  $(-\varepsilon, \varepsilon)$  for some  $\varepsilon > 0$ . Since  $\{x - y \mid x, y \in W_n\} \subseteq f^{-1}(\{x - y \mid x, y \in V\}) \subseteq f^{-1}(U)$ , it follows that  $f^{-1}(U)$  contains  $(-\varepsilon, \varepsilon)$ . Thus  $f$  is continuous at 0. 🧐

**Lemma 1.6.** If  $f : \mathbb{R} \rightarrow \mathbb{R}$  is additive and continuous at one point, then it is continuous on  $\mathbb{R}$ .

**Proof.** Suppose  $f$  is continuous at some  $a \in \mathbb{R}$ . Then we have  $\lim_{x \rightarrow a} f(x) = f(a)$ . For any  $b \in \mathbb{R}$ , we have:

$$\lim_{x \rightarrow b} f(x) = \lim_{x \rightarrow a} f(x - a + b) = \lim_{x \rightarrow a} (f(x) - f(a) + f(b)) = \lim_{x \rightarrow a} f(x) - f(a) + f(b) = f(b)$$

Thus  $f$  is continuous at every  $b \in \mathbb{R}$ . 🧐

**Proposition 1.7.** Suppose  $f : \mathbb{R} \rightarrow \mathbb{R}$  is additive and continuous. Then there is a constant  $a \in \mathbb{R}$  such that  $f(x) = ax$  for all  $x \in \mathbb{R}$ .

**Proof.** Define  $a = f(1)$ . By [Lemma 1.4](#), we have  $f(cy) = cf(y)$  for all  $c \in \mathbb{Q}$  and all  $y \in \mathbb{R}$ . Setting  $y = 1$  yields  $f(c) = cf(1) = ac$ . Suppose  $x \in \mathbb{R}$  and  $(q_n)$  is a sequence in  $\mathbb{Q}$  that converges to  $x$ . Since  $f$  is continuous, the sequence  $(f(q_n))$  converges to  $f(x)$ . This yields  $f(x) = \lim_{n \rightarrow \infty} f(q_n) = \lim_{n \rightarrow \infty} aq_n = a \lim_{n \rightarrow \infty} q_n = ax$ . Thus  $f(x) = ax$  for all  $x \in \mathbb{R}$ . 🧐

**Theorem 1.8.** Suppose  $f : \mathbb{R} \rightarrow \mathbb{R}$  is additive and Lebesgue measurable. Then there is a constant  $a \in \mathbb{R}$  such that  $f(x) = ax$  for all  $x \in \mathbb{R}$ .

**Proof.** By [Proposition 1.5](#),  $f$  is continuous at 0, so by [Lemma 1.6](#), it is continuous on  $\mathbb{R}$ , and so by [Proposition 1.7](#), we have  $f(x) = ax$  for all  $x \in \mathbb{R}$ , where  $a = f(1)$  is a constant. 🧐

**Corollary 1.9.** Suppose  $f : \mathbb{R} \rightarrow \mathbb{R}$  is Lebesgue measurable and  $f(x + y) = f(x)f(y)$  for all  $x, y \in \mathbb{R}$ . Then either  $f \equiv 0$  or there is a constant  $a > 0$  such that  $f(x) = a^x$  for all  $x \in \mathbb{R}$ .

**Proof.** Clearly  $f \equiv 0$  satisfies the above conditions. Note that if  $f(x) = 0$  for some  $x \in \mathbb{R}$ , then  $f(x + y) = 0f(y) = 0$  for all  $y \in \mathbb{R}$ , and so  $f \equiv 0$ . Suppose  $f \not\equiv 0$ . Then  $f(x) \neq 0$  for all  $x \in \mathbb{R}$ . Moreover, we have  $f(x) > 0$  for all  $x \in \mathbb{R}$ , since  $f(x) = f(\frac{x}{2})^2$ . Define  $g : \mathbb{R} \rightarrow \mathbb{R}$  by  $g(x) = \ln(f(x))$ . This is well-defined as  $f(x) > 0$  for all  $x \in \mathbb{R}$ . We also have  $g(x + y) = \ln(f(x + y)) = \ln(f(x)f(y)) = \ln(f(x)) + \ln(f(y)) = g(x) + g(y)$ . Thus  $g$  is additive. It is also Lebesgue measurable as  $g = \ln(f)$  and  $\ln$  is continuous. By [Theorem 1.8](#), we have  $g(x) = bx$  for all  $x \in \mathbb{R}$ , where  $b = g(1)$  is a constant. Thus  $f(x) = e^{bx} = a^x$  for all  $x \in \mathbb{R}$ , where  $a = e^b$ . 🧐

Since  $\mathbb{R}$  is a vector space over  $\mathbb{Q}$ , it has a basis. In other words, there is a set  $\mathcal{B} \subseteq \mathbb{R}$  such that every  $x \in \mathbb{R}$  can be expressed uniquely as  $x = q_1x_1 + q_2x_2 + \dots + q_nx_n$ , where  $x_1, x_2, \dots, x_n \in \mathcal{B}$  and  $q_1, q_2, \dots, q_n \in \mathbb{Q}$ . We now assign arbitrary values to  $f(b)$  for every  $b \in \mathcal{B}$ . Since  $\mathcal{B}$  is linearly independent over  $\mathbb{Q}$ , the value of  $f(b)$  for any given  $b \in \mathcal{B}$  is not affected by the values of  $f$  at other points in  $\mathcal{B}$ , i.e. whatever values  $f$  takes at points in  $\mathcal{B}$ , they will not contradict the equation  $f(x + y) = f(x) + f(y)$ . Also, since  $\mathcal{B}$  spans  $\mathbb{R}$ , the values of  $f$  at points in  $\mathcal{B}$  uniquely determine its values at every point in  $\mathbb{R}$ .

**Definition 1.10.** Suppose  $I \subseteq \mathbb{R}$  is an interval. A function  $f : I \rightarrow \mathbb{R}$  is **midpoint convex** (or *Jensen convex* or *midconvex*) on  $I$  if  $f\left(\frac{x+y}{2}\right) \leq \frac{f(x)+f(y)}{2}$  for all  $x, y \in I$ .

**Remark.** It follows directly (by setting  $\lambda = \frac{1}{2}$  in the definition of a convex function) that every convex function is midpoint convex.

**Theorem 1.11.** If  $f : I \rightarrow \mathbb{R}$  is midpoint convex and continuous, then  $f$  is convex.

**Proof.** Since  $f$  is midpoint convex, for any  $x, y \in I$ , we have  $f\left(\frac{x+y}{2}\right) \leq \frac{f(x)+f(y)}{2}$ . Since  $I$  is an interval, we also have

$\frac{x+y}{2} \in I$ . This implies:

$$f\left(\frac{3}{4}x + \frac{1}{4}y\right) = f\left(\frac{1}{2}\left(\frac{x+y}{2} + y\right)\right) \leq \frac{1}{2}\left(f\left(\frac{x+y}{2}\right) + f(y)\right) \leq \frac{1}{2}\left(\frac{f(x)+f(y)}{2} + f(y)\right) = \frac{3}{4}f(x) + \frac{1}{4}f(y)$$

$$f\left(\frac{1}{4}x + \frac{3}{4}y\right) = f\left(\frac{1}{2}\left(x + \frac{x+y}{2}\right)\right) \leq \frac{1}{2}\left(f(x) + f\left(\frac{x+y}{2}\right)\right) \leq \frac{1}{2}\left(f(x) + \frac{f(x)+f(y)}{2}\right) = \frac{1}{3}f(x) + \frac{1}{4}f(y)$$

Repeating this process, we have for all  $n \in \mathbb{N}$  and all  $m \in \{0, 1, \dots, 2^n\}$ :

$$f\left(\frac{m}{2^n}x + \left(1 - \frac{m}{2^n}\right)y\right) \leq \frac{m}{2^n}f(x) + \left(1 - \frac{m}{2^n}\right)f(y)$$

Now suppose  $x, y \in I$  and  $\lambda \in [0, 1]$ . Then there is a sequence  $(q_n)$  of rational numbers in  $[0, 1]$  with denominators  $2^n$  that converges to  $\lambda$ , e.g.  $q_n = \frac{\lfloor 2^n \lambda \rfloor}{2^n}$ . For each  $n \in \mathbb{N}$ , we have  $f(q_n x + (1 - q_n)y) \leq q_n f(x) + (1 - q_n)f(y)$ . As  $n \rightarrow \infty$ , the right side tends to  $\lambda f(x) + (1 - \lambda)f(y)$ , and since  $f$  is continuous, the left side tends to  $f(\lambda x + (1 - \lambda)y)$ . Thus  $f(\lambda x + (1 - \lambda)y) \leq \lambda f(x) + (1 - \lambda)f(y)$  for all  $x, y \in I$  and all  $\lambda \in [0, 1]$ , and so  $f$  is convex. 🧐

**Proof.** Suppose  $f : I \rightarrow \mathbb{R}$  is midpoint convex and discontinuous at some  $x \in I^\circ$ . Then there is a sequence  $(x_n)$  in  $I$  such that  $\lim_{n \rightarrow \infty} x_n = x$  but  $\lim_{n \rightarrow \infty} f(x_n) \neq f(x)$ . 🧐

**Theorem 1.12.** If  $f : I \rightarrow \mathbb{R}$  is midpoint convex and bounded on some open subinterval  $(a, b) \subseteq I$ , then  $f$  is continuous (and thus convex).

**Theorem 1.13** (Sierpiński's Theorem on Convex Functions). If  $f : I \rightarrow \mathbb{R}$  is midpoint convex and Lebesgue measurable, then  $f$  is convex.

**Theorem 1.14** (Ostrowski's Theorem on Convex Functions). If  $f : I \rightarrow \mathbb{R}$  is midpoint convex and bounded on a Lebesgue measurable set with positive measure, then  $f$  is convex.

**Proof.** Since  $E \in \mathcal{L}(\mathbb{R})$  and  $\mu(E) > 0$ , there is an open set  $G \subseteq I$ ,  $G \supseteq E$  such that  $\mu(E) \leq \mu(G) < \frac{4}{3}\mu(E)$ . Since  $G$  is open, it can be expressed as a countable union of disjoint open intervals  $G = \bigcup_{n=1}^{\infty} I_n$ . Thus we have  $\sum_{n=1}^{\infty} \mu(E \cap I_n) \leq \sum_{n=1}^{\infty} \mu(I_n) < \frac{4}{3}\mu(E \cap I_n)$ . Thus there is at least one  $n \in \mathbb{N}$  such that  $\mu(E \cap I_n) \leq \mu(I_n) < \frac{4}{3}\mu(E \cap I_n)$ . Define  $J = E \cap I_n$ . Since  $f$  is bounded on  $E$ , there exists  $M > 0$  such that  $|f| \leq M$  on  $E$ . Suppose  $f$  is not convex. Then for any  $x, y \in J$ , there exists  $\lambda \in [0, 1]$  such that  $f(\lambda x + (1 - \lambda)y) > \lambda f(x) + (1 - \lambda)f(y)$ . **NOW WHAT?** 🧐

If  $f$  is convex, then  $f$  is differentiable a.e. and its derivative is non-decreasing. Thus  $f$  is twice differentiable a.e.

## 2 Induction

**Theorem 2.1.** For every  $n \in \mathbb{N}$ ,  $\cos(nx)$  and  $\sin(nx)$  can be expressed as polynomials in  $\cos(x)$ .

**Proof.** Define the statement  $P(n)$   $\cos(nx)$  and  $\frac{\sin((n+1)x)}{\sin(x)}$  🧐

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```
from collections.abc import Iterator
```

# This is an example with math. The  $n^{\text{th}}$  Fibonacci number is  $\frac{\phi^n - (1-\phi)^n}{\sqrt{5}}$ , where  $\phi = \frac{1+\sqrt{5}}{2}$  is the golden ratio.

```
class Math:
```

```
    @staticmethod
```

```
    def fib(n: int) -> Iterator[int]:
```

```
        """Fibonacci series up to n."""
```

```
        a, b = 0, 1
```

```
        while a < n:
```

```
            yield a
```

```
            a, b = b, a + b
```

```
result = sum(Math.fib(42))
```

```
print(f"The answer is {result}")
```

**title**

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This is **NOT** the same. Oh, but **NOT**. This is **NOT** \* the same. Oh, but **NOT** \*.

## 2.1

The exercise.

### Solution

The solution.

```
from collections.abc import Iterator

# This is an example
class Math:
    @staticmethod
    def fib(n: int) -> Iterator[int]:
        """Fibonacci series up to n."""
        a, b = 0, 1
        while a < n:
            yield a
            a, b = b, a + b

result = sum(Math.fib(42))
print(f"The answer is {result}")
```

Now another style:

```
print('Did this work?')
```

Also, inline code looks like this.

```
def pascal_triangle(n):
    row = (1,)
    yield row
    for _ in range(n):
        row = (1,) + tuple(map(sum, pairwise(row))) + (1,)
        yield row
```

For example, `list(pascal_triangle(5))` returns:

```
[(1,), (1, 1), (1, 2, 1), (1, 3, 3, 1), (1, 4, 6, 4, 1), (1, 5, 10, 10, 5, 1)]
```

$$\begin{pmatrix} 6 & 5 \\ 4 & 3 \end{pmatrix}$$

$$\lim_{k \rightarrow \infty} f(k) = \lim_{k \rightarrow \infty} f(k) = 0$$

$$\lim_{k \rightarrow \infty} f(k) = \lim_{k \rightarrow \infty} f(k) = 0$$